

HCAHPS Patient Experience Analysis

**Final Executive Presentation
Prepared By: Neha Pandey
Data Analytics Internship
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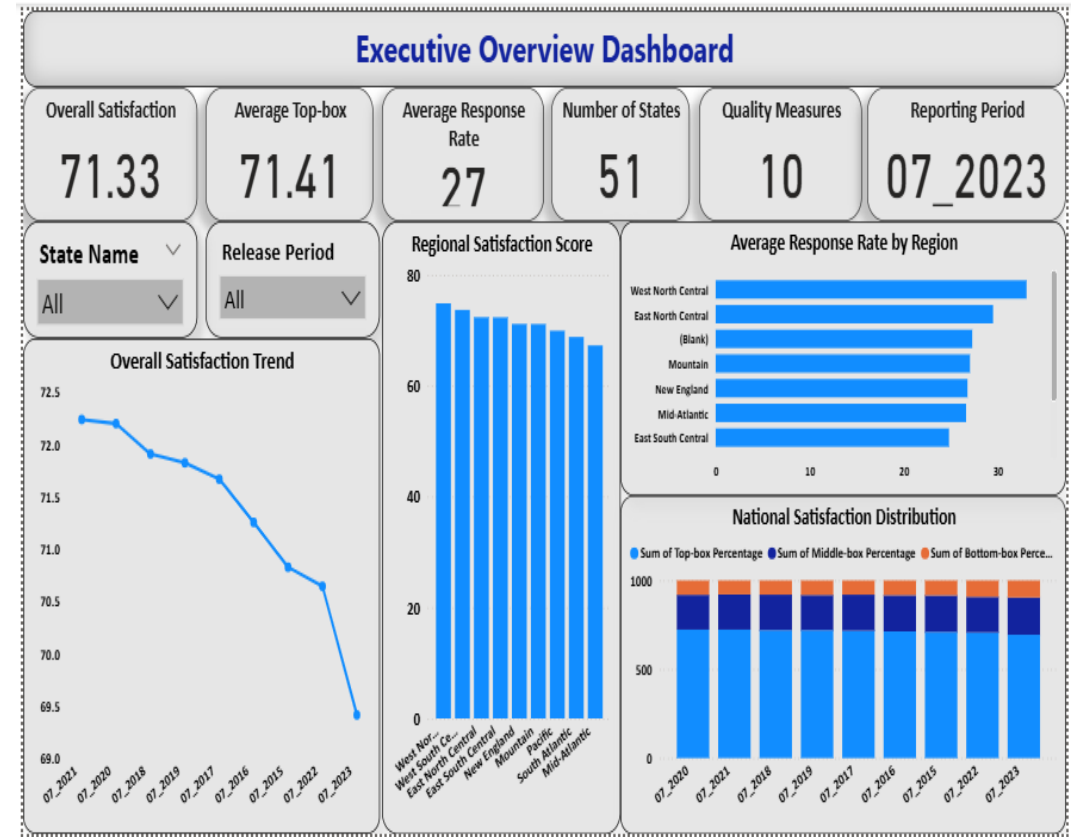
Organization Overview

About HCAHPS

The **Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS)** is a standardized national survey that measures patients' perspectives on hospital care.

The survey helps healthcare organizations to:

- Measure patient satisfaction.
- Evaluate healthcare service quality.
- Compare hospital performance.
- Support data-driven decision-making.
- Improve patient experience across hospitals.



HCAHPS Patient Experience Dashboard



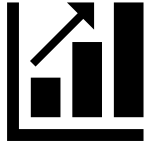
Business Problem



Healthcare organizations face several challenges in monitoring and improving patient satisfaction:

- **Patient satisfaction** varies across hospitals, states, and regions.
- Large volumes of **HCAHPS survey data** make manual analysis difficult.
- Identifying low-performing quality measures requires significant time and effort.
- Hospital administrators need **interactive dashboards** for faster and more accurate decision-making.
- Healthcare organizations require **data-driven insights** to improve **patient experience** and service quality.





Project Objectives

- Analyze **HCAHPS** patient satisfaction data.
- Evaluate hospital performance across states and regions.
- Develop an interactive **Power BI** dashboard.
- Identify trends and performance gaps using data analysis.
- Generate actionable **business insights** and **executive recommendations**.
- Support **data-driven** healthcare decision-making.



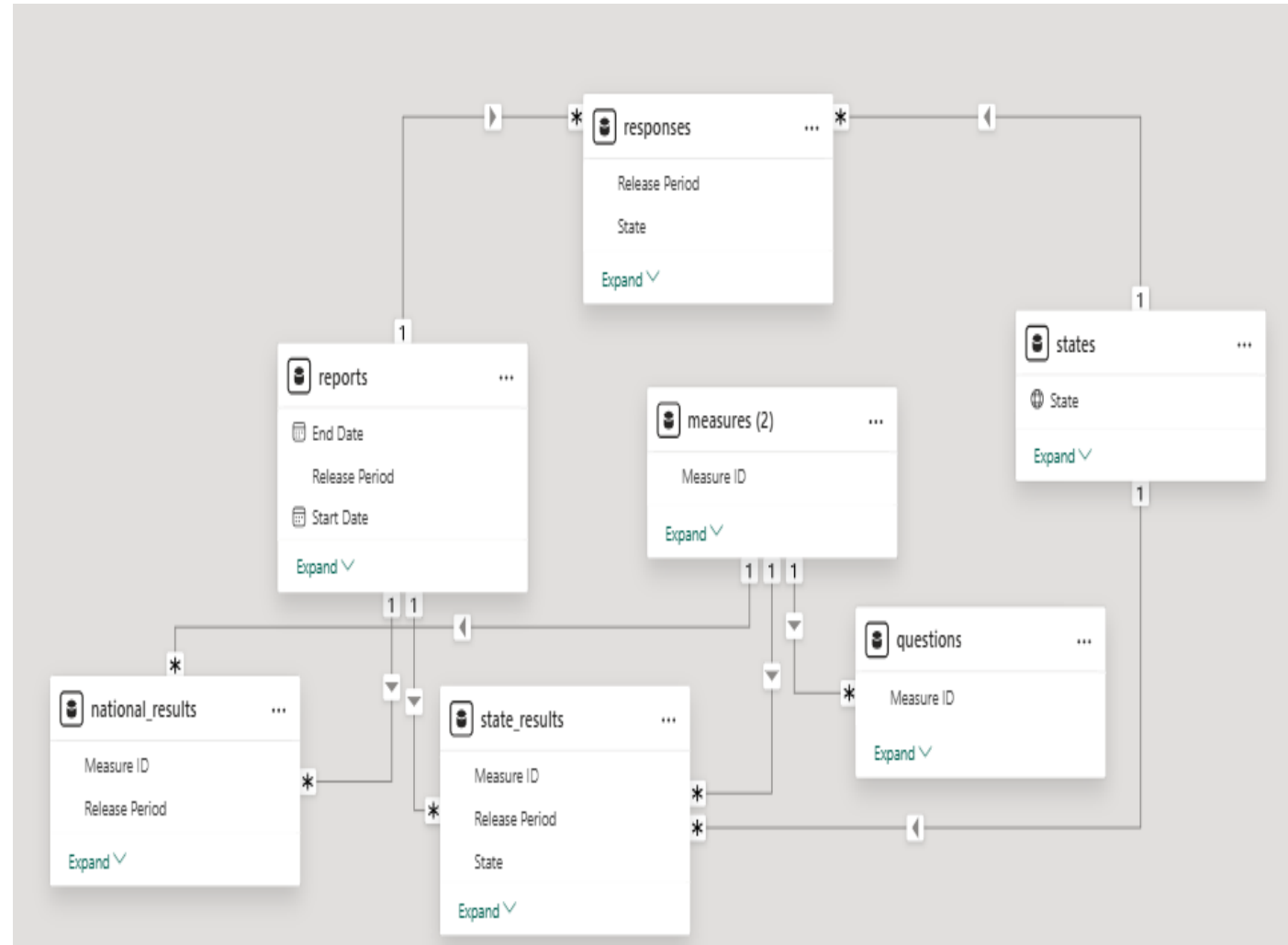
Dataset Overview & Data Model

Dataset Summary

- **Dataset:** HCAHPS Patient Experience Survey
- **Purpose:** Analyze patient satisfaction and hospital performance.
- **Data Source:** CMS (Centers for Medicare & Medicaid Services)
- **Reporting Levels:** National, Regional, and State
- **Reporting Period:** Multiple reporting years
- **Analysis Tool:** Microsoft Power BI

Data Tables

- Measures
- National Results
- State Results
- Responses
- Reports
- Questions
- States



Power BI Relational Data Model (Star Schema)

KPI Framework

Key Performance Indicators (KPIs)

The dashboard monitors the following KPIs to evaluate patient satisfaction and healthcare quality:

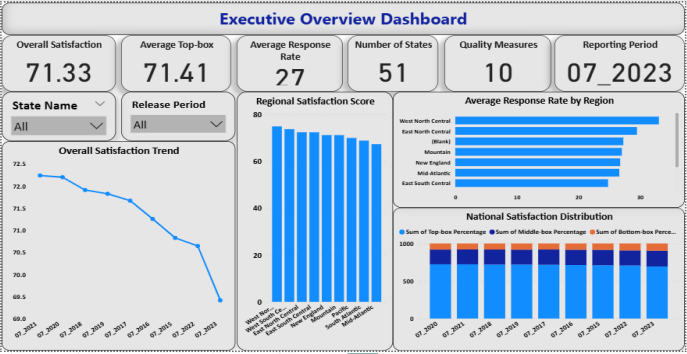
- **Overall Satisfaction Score** – Overall patient satisfaction score.
- **Average Top-box %** – Percentage of patients giving the highest satisfaction rating.
- **Average Response Rate** – Percentage of completed patient surveys.
- **Number of Participating States** – Total states included in the analysis.
- **Number of Quality Measures** – Total healthcare quality measures evaluated.
- **Reporting Period** – Reporting period currently selected for analysis.

Overall Satisfaction	Average Top-box	Average Response Rate	Number of States	Quality Measures	Reporting Period
71.33	71.41	27	51	10	07_2023

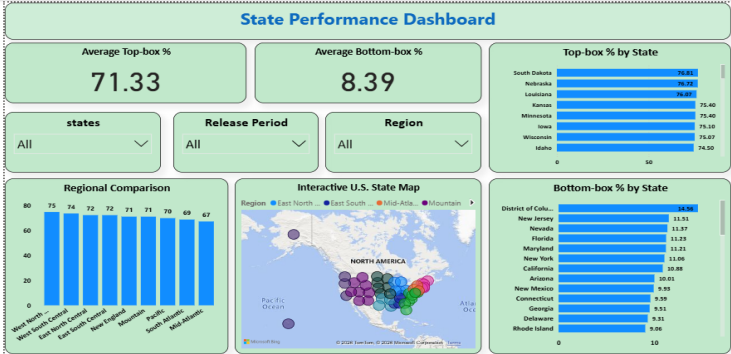
Executive KPI Dashboard



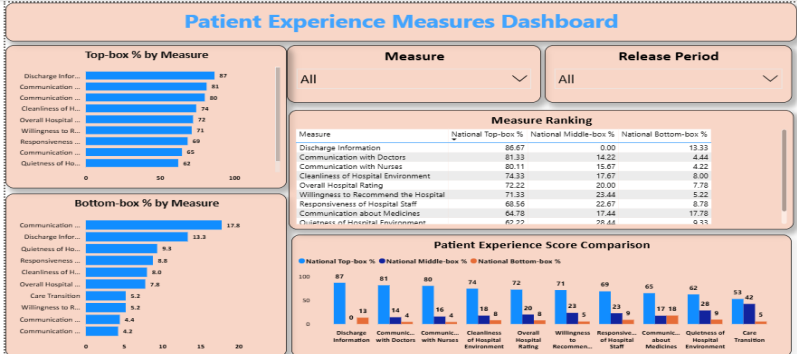
Dashboard Demonstration



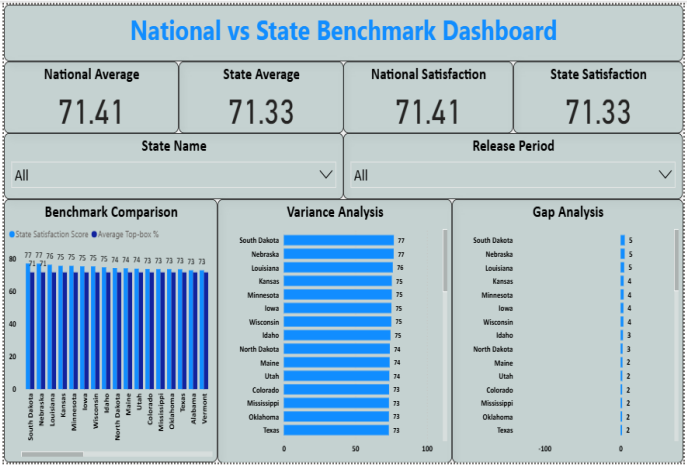
Executive Overview Dashboard



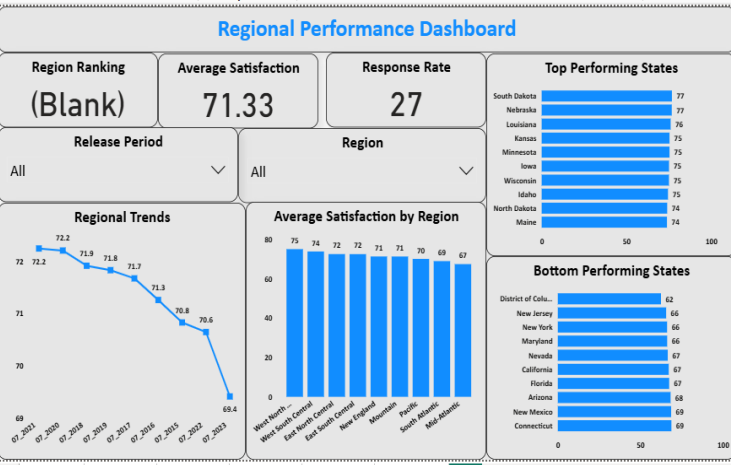
State Performance Dashboard



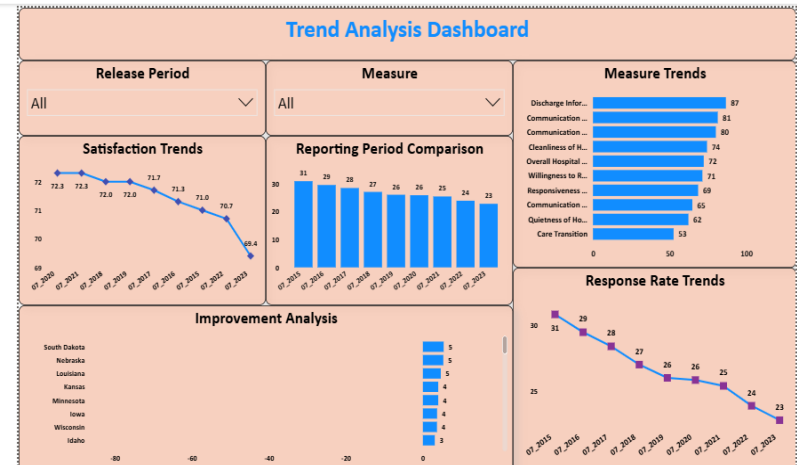
Patient Experience Measures Dashboard



National vs State Benchmark Dashboard



Regional Performance Dashboard



Trend Analysis Dashboard

Key Findings

Major Findings

- Communication with Doctors achieved higher patient satisfaction than Communication about Medicines.
- Patient satisfaction varies across different states and regions.
- Top-performing states consistently maintained higher satisfaction scores.
- Response rates differed across reporting periods, affecting comparison reliability.
- Interactive Power BI dashboards improved healthcare performance monitoring.

Business Impact

- Better communication improves patient trust.
- Regional analysis supports targeted improvements.
- Dashboard insights enable faster decision-making.
- Benchmarking identifies high-performing hospitals.
- Data-driven reporting supports continuous quality improvement.



Strategic Recommendations

Immediate Recommendations

- Improve communication between healthcare providers and patients.
- Conduct regular **staff training** on **patient care and communication**.
- Enhance **patient experience** through quality improvement initiatives.
- Increase HCAHPS **survey participation** and response rates.



Long-Term Recommendations

- **Benchmark** best-performing hospitals and regions.
- Strengthen regional quality improvement programs.
- Support **hospital accreditation** through continuous KPI monitoring.
- Develop **data-driven** healthcare policies using dashboard insights.



Future Roadmap

Future Enhancements

- Integrate **real-time HCAHPS** data for continuous monitoring.
- Apply **Machine Learning** to predict patient satisfaction trends.
- Develop **AI-powered dashboards** for proactive healthcare insights.
- Expand the dashboard to include additional hospital quality metrics.
- Enable **mobile-friendly dashboards** for executives and healthcare managers.
- Automate report generation and KPI monitoring.



Thank You!
Questions & Discussion
Prepared by:
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